Paper 3.75 - Triality Surface as Next-State Precondition

CQE-paper-03.75

CQECMPLX Formal Paper Series

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Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-03.75.md

Paper 3.75 - Triality Surface as Next-State Precondition

Purpose

Paper 3.75 explains how Paper 3 becomes a controlled precondition for Paper 4 and Paper 6.

Exported State

Paper 3 exports:

```
LCR state identity
axis/sheet coordinate
diagonal carrier diag(L,C,R)
trace = shell
trace-2 diagonal idempotent status
Paper 2 correction coordinates (2,0) and (3,1)
```

Use in Paper 4

Paper 4 may consume Paper 3 output as registered boundary-repair input:

```
Paper 2 correction residue + Paper 3 coordinate
-> typed boundary repair constraint
```

The repair paper must preserve the source state, the axis/sheet coordinate, the reason for repair, and the next legal routes. It may not erase the fact that the input came from a correction residue.

Use in Paper 6

Paper 6 generalizes the same discipline from objects to dependencies. Paper 3 registers one state across coordinate languages. Paper 6 registers one proof dependency across typed causal edges.

The bridge is:

```
Paper 3: triality/Jordan registration supplies the registered object.
Paper 6: causal code supplies the registered dependency.
```

The Paper 3 receipt can therefore become a Paper 6 edge receipt, but the edge validity is proved by Paper 6, not by Paper 3.

Use in Later Papers

Later papers may lift the Paper 3 surface into:

- boundary repair constraints,
- rolling path payloads,
- causal proof graphs,
- lattice-code closure frames,
- physics-facing representation claims.

Every lift must state which part of Paper 3 is imported and which later theorem proves the added structure.

Precondition Rule

Before a later paper uses Paper 3, it should be able to answer:

```
Which LCR state is being registered?
What is its axis/sheet coordinate?
What is its diagonal trace?
Is the claim finite registration or stronger group action?
Which receipt proves the imported fact?
```

Conclusion

Paper 3.75 turns the triality surface into a next-state precondition. It makes state registration portable without allowing broader exceptional-algebra language to outrun its receipts.